The QKD Eavesdropping-Detection Pipeline

Simulation, noise models, attacks, features, machine learning and deep learning — in detail

Describes *QuantumKeyDistribution_Draft2.ipynb* (branch claude/wonderful-meitner-16r3p0, including the Draft 2.1 bug-fix commit). Section numbers match the notebook.

1. Overview of the pipeline

Every experiment follows the same flow:

Stage	What happens	Notebook section
1 Simulate	Photon-by-photon (or density-matrix) simulation of one protocol run with a physical noise model and optionally an attacker	1, 2
2 Featurise	Compress a run (millions of pulses) into a short feature vector: error rates, decoy-state estimators, temporal statistics, CHSH values	3
3 Calibrate	Make the three protocols comparable by matching honest QBER, attack-induced excess QBER and number of key rounds	4, 19
4 Generate data	Build labelled datasets (secure vs attacked) with shared channel draws ('common random numbers')	5
5 Audit	Eight leakage tests before any model is trained	6
6 Train & evaluate	Six classical models, ablations, generalisation tests, importance, calibration	7–17, 20
7 Deep learning	Raw event sequences → CNN+BiLSTM+attention; transfer and zero-day experiments	18
8 Operational tests	Minimum detectable strength, CUSUM delay, noise robustness, mixed attacks	21, 22

2. Reproducibility

A single master seed (20260913) feeds a **SeedBook**: the random generator for (role, index) is a numpy SeedSequence built from [master, crc32(role), index]. CRC32 is used because Python's built-in hash() is salted per process. Draft 2.1 fixed a bug where the old class called SeedSequence.spawn(), which is stateful, so the same (role, index) returned different streams on each call; it is now a pure function. A shared constant K_MARGIN = 1.3 sets how much extra pulse budget every equal-information sampler adds.

3. Qubit basics used by the simulators

States are 2-vectors: $|0\rangle$, $|1\rangle$ (Z basis) and $|+\rangle$, $|-\rangle$ (X basis). Measuring in a basis uses the Born rule; measuring a Z state in the X basis (or vice versa) gives a uniformly random bit — the physical reason any basis-guessing attack causes errors. For speed the large simulators exploit that every state is a Z or X eigenstate, so measurement reduces to 'same basis: deterministic, different basis: random'.

4. Noise models

4.1 BB84 and BKM07: fibre channel (Ma, Qi, Zhao & Lo 2005; GYS parameters)

Parameter	Value	Meaning
α	0.21 dB/km	Fibre attenuation at 1550 nm
η_bob	0.045	Bob detector efficiency × optics
Y0	1.7e-6	Dark-count yield per pulse

Parameter	Value	Meaning
e_detector	0.033	Misalignment error of a genuine click
e0	0.5	Error rate of a dark click
f_ec	1.22	Error-correction inefficiency in the key-rate formula

Closed forms: transmittance t = 10^(–αL/10), η = t·η_bob, gain Qμ = Y0 + 1 – e^(–ημ), QBER Eμ = [e0·Y0 + e_detector(1 – e^(–ημ))]/Qμ. QBER is nearly flat with distance (3.30% at 0 km, 3.76% at 100 km) because dark counts only matter when the signal is tiny; the main effect of distance is to cut the number of usable bits. The model is validated only to about 100 km. BB84 uses the channel once; **BKM07 uses it twice** (forward and return leg), so survival is about η^2 (≈0.2% at 0 km) and its distance range is limited to 0–15 km. BKM07 adds a re-preparation error p_prep = e_detector for Bob's classical measure-and-resend.

4.2 BB84 weak coherent pulses and decoy states

Real sources send Poisson-distributed photon numbers. Alice randomly picks one of three intensities — signal μ = 0.48 (70%), decoy 0.05 (25%), vacuum 0 (5%) — so Bob's per-intensity gain and QBER let one bound the single-photon yield Y1 and error e1 (Ma et al. Eqs. 30, 33). The simulator draws all N pulses at once: photon number, bits, bases, loss, dark counts and detector errors.

4.3 E91: Werner state

The shared state is ρ = V|Ψ–⟩⟨Ψ–| + (1–V)I/4. Visibility V sets both the CHSH value |S| = 2√2·V and key QBER (1–V)/2. The honest draw is V ~ Uniform(0.85, 0.99). Observables A(θ) = cos2θ·σz + sin2θ·σx with Alice at 0°, 22.5°, 45° and Bob at 22.5°, 45°, 67.5°. Pairs (22.5,22.5) and (45,45) are anti-correlated and give key bits; four other pairs form S = E(a1,b1) – E(a1,b3) + E(a3,b1) + E(a3,b3), classical bound 2, quantum bound 2.83. Outcomes are sampled exactly from Born-rule probabilities Tr[ρ(Π⊗Π)], 9 setting pairs × honest/attacked states.

5. Attack models

When Eve acts is a separate process: *iid* (each pulse attacked with probability = intensity), *bursty* (exponentially distributed on/off bursts) or *drifting* (sinusoidally modulated rate). Without temporal structure, the temporal features would have nothing to measure.

Protocol	Attack	Mechanism	Visible as
BB84	Intercept-resend	Eve measures in a random basis and resends; wrong basis (50%) randomises the bit	Sifted QBER up (0.5 on those pulses, ~0.25 average at full intensity)
BB84	PNS	Eve blocks n<2 photon pulses, splits one photon from n≥2 pulses and forwards the rest losslessly, throttled so Bob's gain looks honest	No extra QBER; distorts gain ratio between decoy and signal intensities
BB84	Blocking / loss manipulation	Eve blocks pulses or attenuates and boosts the rest	Click-rate changes (only used in DL data)
BB84	Mixed PNS+IR	Both at once on the same schedule	Both signatures (disjoint outputs: error vs yield)
BKM07	Forward / return intercept	With probability eve_fwd Eve measures the forward photon in a random basis; with eve_ret the returning photon. 'Symmetric' = equal, 'asymmetric' = weak forward, strong return	CTRL-round error rate (reflected photon should return unchanged)
E91	Intercept-resend	Basis-averaged measure-and-resend CPTP map on Bob's arm	QBER up, |S| falls toward 2
E91	Entangling ancilla	Anisotropic dephasing (Fuchs-type probe)	Costs more QBER than CHSH: s_qber_residual
E91	Loss manipulation	Asymmetric arm attenuation modelled as extra depolarisation	Looks like honest visibility loss

6. BKM07 round structure

Alice prepares |0⟩,|1⟩,|+⟩,|–⟩. Classical Bob either **SIFT**s (measure in Z, re-prepare, send back) or **CTRL**s (reflect unchanged), each with probability 0.5. Alice measures the return in her own preparation basis. Round types: SIFT_KEY (Alice sent Z, Bob sifted: the key), SIFT_MONITOR (Alice X + SIFT: discarded, used only as a return-leg monitor), CTRL_Z and CTRL_X (reflection checks that expose Eve). Honest end-to-end key QBER is three composed binary-symmetric channels: (1 – (1–2e)³)/2.

7. Feature extraction (Section 3)

7.1 Temporal helpers

Runs are cut into W windows (e.g. 64) and per-window error counts and sifted counts kept. **Dispersion index** D = Var(Qˆ)·n■ / [Q■(1–Q■)] is the observed variance relative to the binomial floor: ~1 for a stationary process, >1 if the true error rate varies in time (replaces raw variance, which duplicates the mean). **Normalised jump energy** is the sum of squared consecutive window differences divided by its stationary expectation. **Spectral entropy** is the Shannon entropy of the window-QBER power spectrum (0 = structured, 1 = noise-like; biased low for few windows). **Lag-1 autocorrelation** detects clustering.

7.2 Feature sets

Protocol	Features (count)
BB84 (14)	qber_total, qber_z, qber_x, gain_mu, sifted_rate, qber_dispersion, jump_energy_norm, spectral_entropy, autocorr_lag1, y1_lower, e1_upper, r_secure, gain_ratio_nu_mu, h_qber
BKM07 (14)	qber_key (SIFT_KEY end-to-end error), qber_zs, qber_zsr, ret_monitor, qber_zc, qber_xc, asymmetry, ctrl_sift_ratio, xctrl_dispersion, jump_energy, spectral_entropy, autocorr_lag1, h_ctrl, sifted_rate
E91 (11)	qber_key, chsh_S, s_deviation, s_qber_residual, per_pair_corr_spread, jump_energy (of per-window CHSH), spectral_entropy, autocorr_lag1, h_qber_key, sifted_rate, r_secure_di

Decoy estimators: Y1 lower bound and e1 upper bound from signal, decoy and vacuum gains/QBERs; **r_secure** = q·{–Qμf_ec H(Eμ) + Q1[1 – H(e1)]} (GLLP + decoy asymptotic rate, used as a feature, not a security proof); **gain_ratio_nu_mu** = Qνe^ν / Qμe^μ, which is the same for honest links but is broken by PNS because Eve cannot tell decoy from signal pulses. **s_qber_residual** = |S| – 2√2(1 – 2Q): zero for honest depolarisation, positive for the anisotropic ancilla. **r_secure_di** is the Acín et al. device-independent bound 1 – h(Q) – h((1+√((S/2)²–1))/2). Removed on purpose: the unobservable true multi-photon rate and features that were deterministic functions of QBER (raw variance).

8. Equal-information sampling and calibration

Fixed pulse counts give wildly different numbers of usable bits per protocol and distance. The collectors therefore take a target K of key events (2,000 in the main datasets) and size N from the closed-form rate: BB84 N = K/(0.7·gain·0.5)·margin; BKM07 N = K/(η²·0.25)·margin; E91 N = K/(2/9)·margin. BB84 is capped at 20M pulses and BKM07 at 15M, which Draft 2.1 now reports and uses to cap K for all protocols in cross-protocol benchmarks.

Calibration (Section 4): honest noise is inverted in closed form — BB84 solves distance for a target QBER, E91 V = 1 – 2Q, BKM07 e_detector from the three-BSC formula — and attack strength is solved numerically (Monte-Carlo plus Brent root-finding on a fixed seed) to produce a chosen excess QBER. This lets the three protocols be compared at the same honest and attacked error rates.

9. Dataset generation (Section 5)

For each run index i the nuisance parameters (BB84: distance U(0,100) km, e_detector ~ N(0.033, 0.005), Y0 log-uniform 1e-6..1e-5; BKM07: distance U(0,15)) are drawn once and reused for that index's secure and attacked samples ('common random numbers'). Datasets: BB84 900 runs (none / intercept-resend / PNS, 300

each), BKM07 900 runs (none / symmetric / asymmetric), E91 1,200 runs (none / intercept-resend / ancilla / loss manipulation, 300 each; V ~ U(0.85,0.99), intensity U(0.1,0.9)). Labels are binarised (0 secure, 1 attacked) for the detectors. Because rows come in channel-matched triplets, **every split is by run index** (GroupShuffleSplit / StratifiedGroupKFold) so a clean run and its attacked twin never straddle train and test.

10. Leakage audit (Section 6)

Following Kapoor & Narayanan (2023): L-1 single-feature AUC > 0.99; L-2/L-3 constant or disjoint features; L-4 KS test of nuisance parameters across classes; L-5 identical sample length across classes; L-6 NaN-rate differences; L-7 shuffled labels must give AUC ~0.5; L-8 the **zero-strength control** keeps the class labels but sets all attack strengths to zero, so AUC must be ~0.5 — the strongest test that the label is not leaking through the simulator. L-8 is run for all three protocols in Draft 2.1.

11. Classical machine learning (Sections 7–17, 20)

- **Models**: KNN (k=7), logistic regression, random forest, SVM-RBF, XGBoost (HistGradientBoosting fallback), and an Isolation Forest fitted only on secure data (unsupervised, score = anomaly). All supervised models sit behind a StandardScaler pipeline so scaling is learned on training data only.
- **Tuning**: grid search maximising ROC-AUC with 5-fold *group-aware* stratified CV (grids: RF 27 combinations, SVM 16, boosted trees 54).
- **Metrics**: accuracy, AUC, F1; FNR/FPR for security; FNR at a fixed 1% false-positive rate with the threshold chosen on a held-out validation slice of the training data (not the test data).
- **Ablation A / A+ / B / C**: QBER only → + aggregates → + temporal → + decoy, 20 repeated group splits with 95% Student-t intervals and re-seeding under 3 master seeds.
- **Generalisation E1–E4**: random split; train weak/test strong intercept-resend and the reverse; train near/test far distance and reverse; leave-one-(distance×intensity)-cell-out.
- **Importance**: permutation importance on held-out data (impurity importance is biased), plus SHAP TreeExplainer. Feature groups are correlated, so read rankings by group.
- **Sample-size study** (Section 16): AUC of QBER-only vs QBER+aggregate+temporal across $N \in \{2e5,1e6,2e6\}$ and $W \in \{16,32,64\}$.
- **E91 extras**: CHSH-only baseline vs full 11 features, FNR per attack, train weak/test strong duty cycle.
- **Calibration & thresholds** (Section 20): Platt-scaled CalibratedClassifierCV (sigmoid, as data are few hundred rows), reliability curves, Brier score, operating thresholds at 1/5/10% FPR, and a closed-form base-rate correction posterior odds = likelihood ratio × deployment prior odds to show how precision collapses when attacks are rare.

12. Deep learning (Section 18)

12.1 Input encoding

Each session targets 2,000 informative **events** (a BB84 click, a BKM07 surviving round trip, an E91 pair) rather than raw time slots, because raw slots are mostly empty. Eight channels per event:

Channel	BB84	BKM07	E91
kept	sifted click	SIFT_KEY round	key-setting pair
error	kept & Bob≠Alice	error suited to round type	kept & not anti-correlated
basis_a / basis_b	Alice / Bob basis	Alice basis / Z or n.a.	angle setting a1..a3 / b1..b3
test_flag	decoy/vacuum pulse	CTRL round	one of the 4 CHSH pairs

Channel	BB84	BKM07	E91
event_gap	log of raw pulses since last click	since last surviving trip	~constant (no loss)
aux1	intensity class	round-type code	CHSH index
chsh_running	0	0	rolling CHSH / 2√2

Basis values are normalised to [0,1]; event_gap is log-compressed and clamped. Sessions are cut into overlapping windows of 96 events (stride 48); sessions shorter than 96 events are dropped rather than zero-padded. Dataset: 25 sessions per class, 11,483 windows (BB84 4,483, E91 4,000, BKM07 3,000). Splits are always by **session**, stratified by label (Draft 2.1).

12.2 Architecture (CrossProtocolDetector)

- **Per-protocol adapter**: a linear layer 8 → 48 for each protocol.
- **Shared trunk**: three dilated Conv1d layers (hidden 64, kernels 5/5/3, dilations 1/2/4, BatchNorm, ReLU, dropout 0.2) → 2-layer bidirectional LSTM (hidden 64) → **attention pooling** (learned softmax weights over the 96 steps) → 128-d embedding.
- **Heads**: a shared binary head (attacked vs not), protocol-specific multi-class heads for the attack type, and a protocol discriminator.
- **DANN**: the discriminator is fed through a gradient-reversal layer (λ ramped 0→1) to push the trunk toward protocol-invariant embeddings. **SupCon**: supervised-contrastive loss on the protocol-mixed batch pulls 'attacked' embeddings of different protocols together.
- **Training**: AdamW (lr 1e-3, weight decay 1e-4), cosine schedule, gradient clipping 1.0, checkpoint chosen by *validation AUC* (F1 is gamed by predicting 'always attacked' when 67–80% of windows are attacked).

12.3 Experiments

- **Leave-one-protocol-out transfer**: pretrain on two protocols (adversarial + contrastive), freeze the trunk (conv, LSTM, attention in eval mode so BatchNorm statistics do not change), train a fresh adapter (and the small shared binary head) on 5–100% of the target protocol's sessions; compare with the same architecture trained from scratch on the same data; 5 seeds; paired bootstrap CI; plus a classical boosted-tree baseline on the physics features at matched K.
- **Deep SVDD zero-day test**: hold one attack type out of all training. Branch A: supervised representation (frozen) with the centre of clean embeddings; Branch B: a bias-free encoder (no biases, non-affine BatchNorm, mean pooling, to prevent hypersphere collapse) trained to pull clean embeddings toward a fixed centre. Score = squared distance from the centre; evaluated as AUC on clean vs held-out attack.
- **Grouped 5-fold CV** pooled across protocols (3 seeds) for full coverage, and an **embedding PCA** diagnostic.
- **LSTM autoencoder** is defined but not used in the main flow.

13. Operational security experiments (Sections 21–22)

- **Minimum detectable strength**: sweep the raw attack knob (BB84 eve_intensity, BKM07 eve_fwd = eve_ret, E91 ancilla λ) at a fixed calibrated honest QBER and report the smallest strength with AUC ≥ 0.8.
- **CUSUM** (Page 1954): a log-likelihood-ratio sequential test on the Bernoulli error indicator of key rounds only. Honest baseline estimated from a 300-key-round burn-in; detector shift fixed at +0.05, threshold 5 nats; 600 attacked key rounds follow; the delay (alarm index – change-point) is the number of key rounds that leak before the alarm.
- **Bursty-but-honest**: honest sessions at 1–5× the calibrated noise are fed to the trained classifiers to measure false-positive rate.

- **Mixed attack**: intercept-resend + PNS simultaneously vs each alone.

14. Bug-fix pass (Draft 2.1)

A code review found and fixed: a non-deterministic seed generator; a CUSUM experiment starved of key rounds for BB84/BKM07; a hardcoded sample length in the E91 audit; unequal key budgets across protocols (shared K_MARGIN, reported pulse caps, matched-K benchmarks); a DL classical baseline with ~5 key rounds; twin-row leakage in hyper-parameter CV, Section 20 and E2; thresholds chosen on test data; overlapping re-seed splits; unstratified DL session splits; a trunk that was not truly frozen; pseudo-replicated significance tests; and plotting/installation hygiene. Items not resolved: BB84 deep learning at chance, saturated benchmarks (AUC 1.0), and the trivially separable BKM07 dataset.

15. Limitations

- Individual per-pulse attacks only; no collective/coherent attacks, detector blinding or Trojan-horse attacks.
- r_secure and the DI rate are asymptotic features, not finite-key security bounds. A BKM07 key-rate bound was not implemented.
- Channel model validated only to about 100 km; BKM07 only to 15 km; E91 has no loss model.
- Everything is simulated; no real QKD traces were tested.
- Many experiments saturate at AUC 1.0, so they cannot discriminate between protocols or models.